

# global functions

```
#ifndef CSMP_BOUNDARY_CONDITIONS_H
#define CSMP_BOUNDARY_CONDITIONS_H

#include "CSMP_definitions.h"
#include "Box.h"
#include "Point.h"

namespace csmp {
/**
@file BoundaryConditions.h
@brief Global boundary functions
@author S.K. Matthaei
*/

// boundary conditions
bool isEdge( BOX_BOUNDARY );
bool isCorner( BOX_BOUNDARY );
bool isSide( BOX_BOUNDARY );
bool belongsToEdge( BOX_BOUNDARY edge, BOX_BOUNDARY );
bool belongsToSide( BOX_BOUNDARY side, BOX_BOUNDARY );
bool isLEFT( BOX_BOUNDARY );
bool isRIGHT( BOX_BOUNDARY );
bool isTOP( BOX_BOUNDARY );
bool isBOTTOM( BOX_BOUNDARY );
bool isFRONT( BOX_BOUNDARY );
bool isBACK( BOX_BOUNDARY );

void boundaryMinMaxCoordinates( BOX_BOUNDARY,
                               csmp::Point<1U>& model_coord_min,
                               csmp::Point<1U>& model_coord_max );

void boundaryMinMaxCoordinates( BOX_BOUNDARY, csmp::Point<2U>&, csmp::Point<2U>& );
void boundaryMinMaxCoordinates( BOX_BOUNDARY, csmp::Point<3U>&, csmp::Point<3U>& );

/**
@addtogroup CSMPglobalFunctions
@{
*/

// inline functions

inline bool isEdge( BOX_BOUNDARY bd )
{
    if ( bd == EDGE1 ) return true;
    if ( bd == EDGE2 ) return true;
    if ( bd == EDGE3 ) return true;
    if ( bd == EDGE4 ) return true;
    if ( bd == EDGE5 ) return true;
    if ( bd == EDGE6 ) return true;
    if ( bd == EDGE7 ) return true;
    if ( bd == EDGE8 ) return true;
    ...
}
```

```
inline bool isBACK( BOX_BOUNDARY bd )
{
    if ( bd == BACK ) return true;
    if ( bd == EDGE1 ) return true;
    if ( bd == EDGE2 ) return true;
    if ( bd == EDGE3 ) return true;
    if ( bd == EDGE4 ) return true;
    if ( bd == CNR1 ) return true;
    if ( bd == CNR2 ) return true;
    if ( bd == CNR3 ) return true;
    if ( bd == CNR4 ) return true;
    return false;
}

/**
@}
*/

} // end csmp

#endif
```

csmp::CNR4

# global enums

```
#ifndef CSMP_BOX_H
#define CSMP_BOX_H

#include <iostream>
#include <vector>
#include "CSMP_number_types.h"

namespace csmp {
/**
@file Box.h
@brief definitions for rectangular and box-shaped models
@author S.K. Matthaei
*/

/**
@addtogroup CSMPglobalEnums
@{
*/
enum { IRREGULAR_OUTSIDE = -1,
      LEFT_OUTSIDE      = -2, // flagging the model boundary
      RIGHT_OUTSIDE     = -3,
      BOTTOM_OUTSIDE    = -4,
      TOP_OUTSIDE       = -5,
      FRONT_OUTSIDE     = -6,
      BACK_OUTSIDE      = -7,
      CNR_MIN           = -8,
      CNR_MAX           = -9,
      CNR_MIN_MAXX      = -10,
      CNR_MIN_MAXXZ     = -11,
      CNR_MIN_MAXZ      = -12,
      CNR_MAX_MINXZ     = -13,
      CNR_MAX_MAXX      = -14,
      CNR_MAX_MAXZ      = -15,
      BACK_BOTTOM       = -16, // BACK and BOTTOM
      BACK_RIGHT        = -17, // BACK and RIGHT
      BACK_TOP          = -18, // BACK and TOP
      BACK_LEFT         = -19, // BACK and LEFT
      BOTTOM_RIGHT       = -20, // BOTTOM and RIGHT
      TOP_RIGHT         = -21, // TOP and RIGHT
      TOP_LEFT          = -22, // TOP and LEFT
      BOTTOM_LEFT        = -23, // BOTTOM and LEFT
      FRONT_BOTTOM      = -24, // FRONT and BOTTOM
      FRONT_RIGHT       = -25, // FRONT and RIGHT
      FRONT_TOP         = -26, // FRONT and TOP
      FRONT_LEFT        = -27, // FRONT and LEFT
      REGION_BOUNDARY   = -28
};
```